

Class 10: Structural Bioinformatics

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1. PBB Statistics

The main repository of biomolecular structure data is called the [Protein Data Bank](#) (PDB for short). It is the second oldest database (after GenBank).

What is currently in the PDB? We can access current composition stats [here](#)

```
stats <- read.csv("Data Export Summary.csv")
stats
```

	Molecular.Type	X.ray	EM	NMR	Multiple.methods	Neutron	Other
1	Protein (only)	171,959	18,083	12,622	210	84	32
2	Protein/Oligosaccharide	10,018	2,968	34	10	2	0
3	Protein/NA	8,847	5,376	286	7	0	0
4	Nucleic acid (only)	2,947	185	1,535	14	3	1
5	Other	170	10	33	0	0	0
6	Oligosaccharide (only)	11	0	6	1	0	4
	Total						
1	202,990						
2	13,032						
3	14,516						
4	4,685						
5	213						
6	22						

Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

```
#substitute commas for nothing and convert to numeric
xray.num <- as.numeric(gsub(",", "", stats$X.ray))

#sum the result
sum(xray.num)
```

```
[1] 193952
```

Turn this snippet into a function so I can use it any time I have this comma problem.

```
comma.sum <- function(x) {
  y <- sum(as.numeric(gsub(",", "", x)))
  return(y)
}
```

```
xray.sum <- comma.sum(stats$X.ray)
em.sum <- comma.sum(stats$EM)
total.sum <- comma.sum(stats$Total)

xray.sum/total.sum * 100
```

```
[1] 82.37223
```

```
em.sum/total.sum * 100
```

```
[1] 11.30648
```

Q2: What proportion of structures in the PDB are protein?

```
#includes Oligosaccharides and NA values
total.num <- as.numeric(gsub(",", "", stats$Total))
sum(total.num[1:3])/total.sum
```

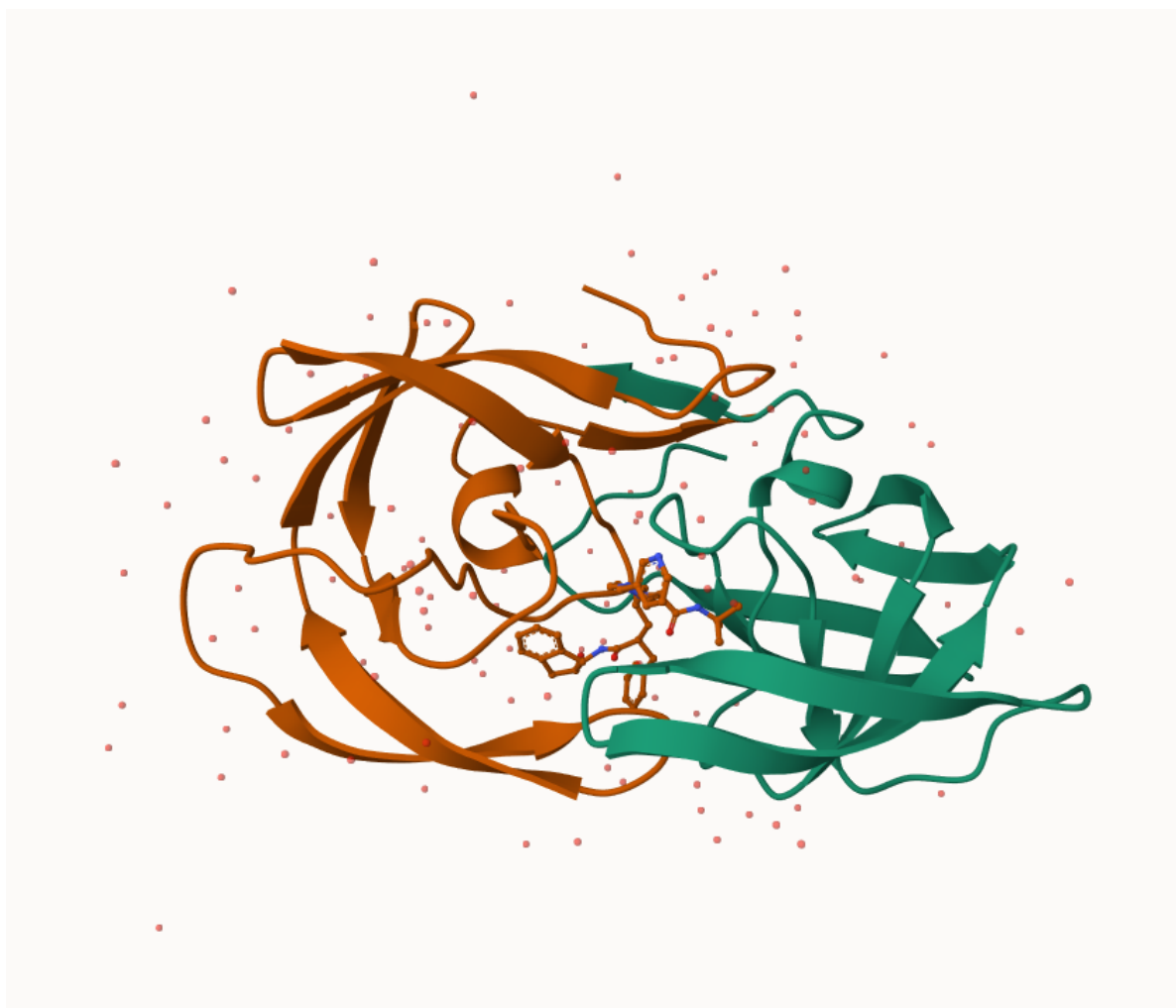
```
[1] 0.9791046
```

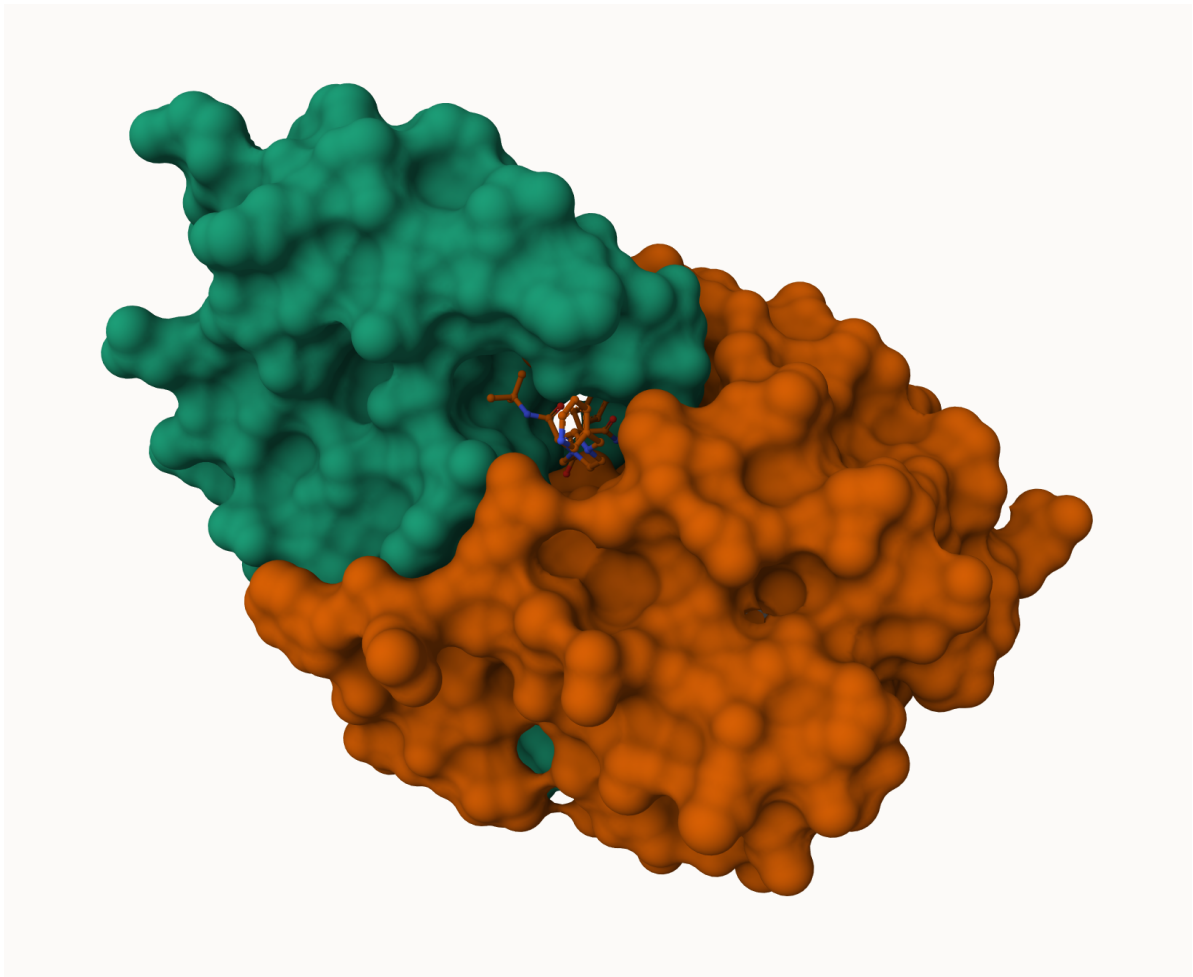
Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

SKIPPED

2. Visualizing with Mol-star

Explore the HIV-1 protease structure with PDB code: 1HSG using [Molstar](#)





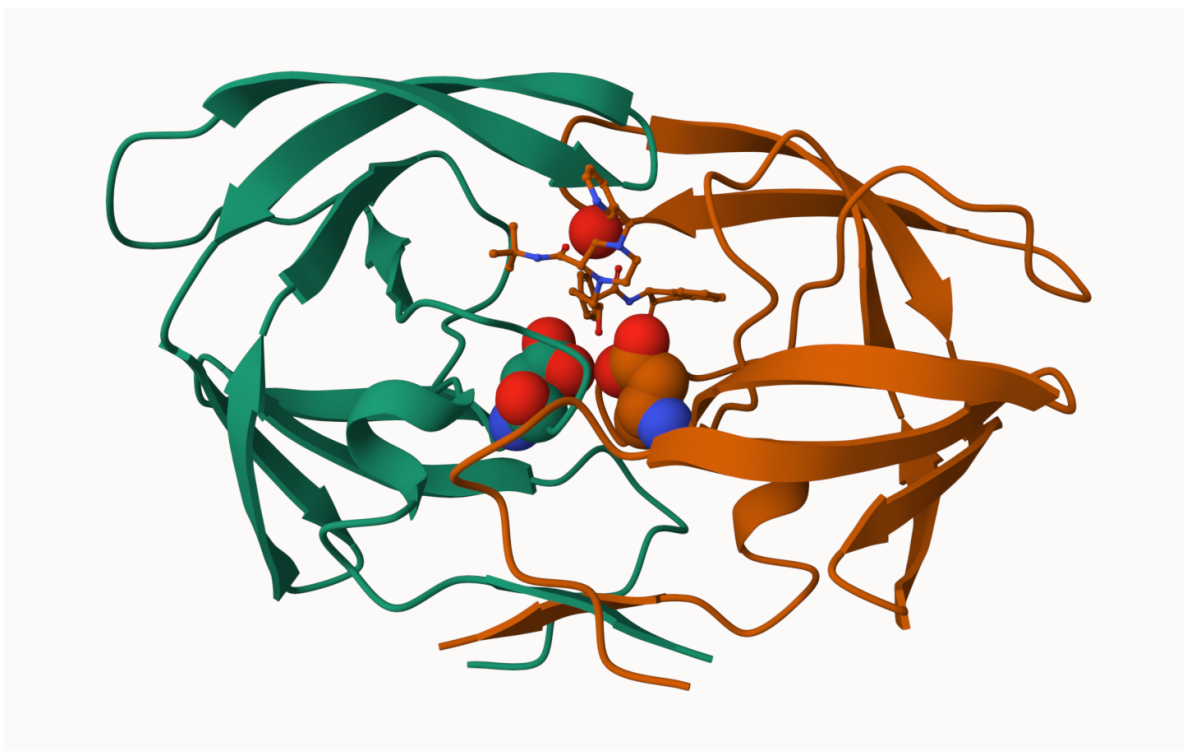


Figure 1: Figure 3: The catalytically important ASP 25 and drug interacting HOH 308 water molecule

3. Using the2 Bio3D in R

The bio3d package is used for bioinformatics analysis and allows us to read and analyze PDB (and related) data.

```
library(bio3d)
```

```
pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
pdb
```

```
Call: read.pdb(file = "1hsg")
```

```

Total Models#: 1
Total Atoms#: 1686, XYZs#: 5058 Chains#: 2 (values: A B)

Protein Atoms#: 1514 (residues/Calpha atoms#: 198)
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 172 (residues: 128)
Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]

```

```

Protein sequence:
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
VNIIGRNLLTQIGCTLNF

```

```

+ attr: atom, xyz, seqres, helix, sheet,
      calpha, remark, call

```

```
head(pdb$atom)
```

	type	eleno	elety	alt	resid	chain	resno	insert	x	y	z	o	b
1	ATOM	1	N	<NA>	PRO	A	1	<NA>	29.361	39.686	5.862	1	38.10
2	ATOM	2	CA	<NA>	PRO	A	1	<NA>	30.307	38.663	5.319	1	40.62
3	ATOM	3	C	<NA>	PRO	A	1	<NA>	29.760	38.071	4.022	1	42.64
4	ATOM	4	O	<NA>	PRO	A	1	<NA>	28.600	38.302	3.676	1	43.40
5	ATOM	5	CB	<NA>	PRO	A	1	<NA>	30.508	37.541	6.342	1	37.87
6	ATOM	6	CG	<NA>	PRO	A	1	<NA>	29.296	37.591	7.162	1	38.40

	segid	elesy	charge
1	<NA>	N	<NA>
2	<NA>	C	<NA>
3	<NA>	C	<NA>
4	<NA>	O	<NA>
5	<NA>	C	<NA>
6	<NA>	C	<NA>

Q7: How many amino acid residues are there in this pdb object?

198

Q8: Name one of the two non-protein residues?

HOH

Q9: How many protein chains are in this structure?

2

```
library(bio3dview)
library(NGLVieweR)

#view.pdb(pdb, backgroundColor = "pink", colorScheme = "sse") |>
#setSpin()

# Select the important ASP 25 residue
sele <- atom.select(pdb, resno=25)

# and highlight them in spacefill representation
#view.pdb(pdb, cols=c("green","orange"), highlight = sele, highlight.style = "spacefill") |>
#setRock()
```

Predicting functional motions of a single structure

We can finish off today with a bioinformatics prediction of the functional motions of a protein.

We will run a Normal mode Analysis (NMA)

```
adk <- read.pdb("6s36")
```

Note: Accessing on-line PDB file

PDB has ALT records, taking A only, rm.alt=TRUE

```
adk
```

Call: read.pdb(file = "6s36")

Total Models#: 1

Total Atoms#: 1898, XYZs#: 5694 Chains#: 1 (values: A)

Protein Atoms#: 1654 (residues/Calpha atoms#: 214)

Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 244 (residues: 244)

Non-protein/nucleic resid values: [CL (3), HOH (238), MG (2), NA (1)]

Protein sequence:

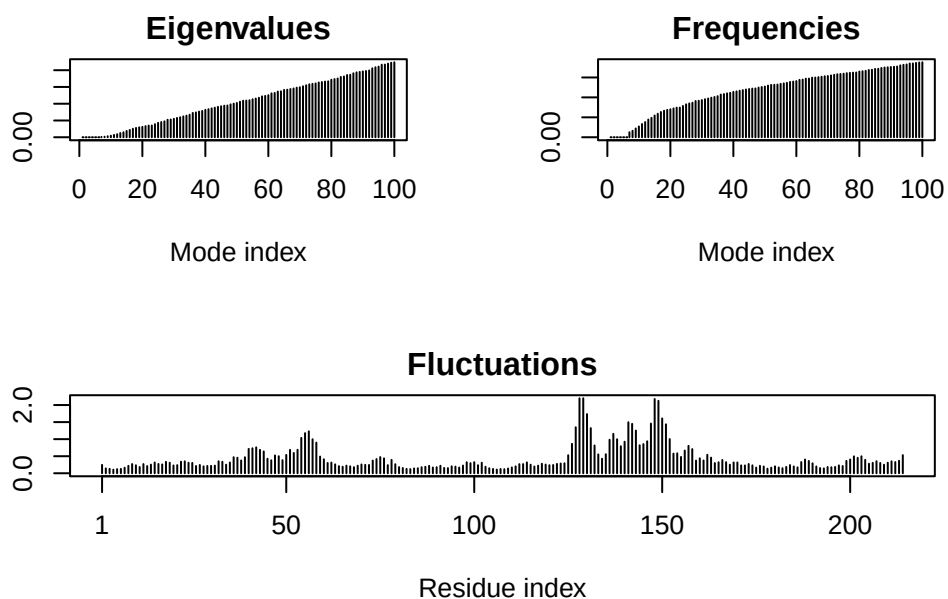
```
MRILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLRAAVKSGSELGKQAKDIMDAGKLV  
DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDKI  
VGRRVHAPSGRVYHVKFNPVKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG  
YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
```

```
+ attr: atom, xyz, seqres, helix, sheet,  
      calpha, remark, call
```

```
m <- nma(adk)
```

```
Building Hessian...      Done in 0.044 seconds.  
Diagonalizing Hessian... Done in 0.492 seconds.
```

```
plot(m)
```



```
mktrj(m, file="adk_m7.pdb")  
#view.nma(m, pdb=adk)
```